

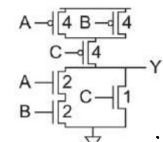
halo注入 D区大掺杂浓度

- 二阶效应 ①电荷共享 $I_{D1} \Rightarrow V_{T1} \downarrow$ 电场耗尽电荷 ②短沟效应 $L \downarrow V_{T1} \downarrow$ ③DDBL $V_{T1}' = V_{T1} - qV_{DS} V_{DS} \uparrow$ 耗尽区 $V_{T1} \downarrow V_{DS} \uparrow$ 源漏漏通
- ④反窄沟效应 W 小时 V_G 维持额外电场耗尽电荷 $\Rightarrow$ 浅槽隔离技术. 预先使隔离处边缘电场 $E \uparrow \Rightarrow V_{T1} \downarrow$ 耗尽区耗尽层在两侧向沟内
- 漏电流 ①亚阈值漏电流 $V_{GS} < V_T$ $I_D \sim I_0 e^{\frac{qV_{GS}}{nkT}} = I_0 e^{\frac{V_{GS}}{10}}$ $S = n \frac{kT}{q} \ln(10) \sim 60-100 mV/decade$ $n = \frac{C_{ox}}{C_{ox}} + 1 \sim 1.2-1.5$
- $I_D = I_0 \exp((V_{GS} - V_T)/(nkT/q)) (1 - \exp(-V_{DS}/(kT/q)))$ $I_0 \propto K^{\frac{W}{L}}$
- ②GIDL $V_{GS} \leq 0$ 漏端电压高 G 和 D 之间强电场引起价带隧穿. 流入漏极. 产生空穴流入衬底 $t_{ox} \downarrow I_d \uparrow V_{GS} \uparrow I_d \uparrow$
- ③栅氧漏电流 $I_g = WA (\frac{V_{GS}}{t_{ox}})^2 e^{-D \frac{qV_{GS}}{kT}}$ ④PN结漏电流 $I_{DL} = J_s A (e^{\frac{V_{DS}}{V_{T0}}} - 1)$
- 温度对迁移率影响 $\mu(T) = \mu(T_r) (\frac{T_r}{T})^{-k_\mu}$ $k_\mu = 1.2 \sim 2.0$ T_r 室温 $\Rightarrow V_T(T) = V_T(T_r) - k_{VT}(T - T_r)$ $k_{VT} = 0.5 \sim 1 mV/K$
- $T \uparrow \Rightarrow \mu \downarrow V_T \downarrow I_{DS} \downarrow I_D$ 亚阈值 $\uparrow \Rightarrow$ 导通 $T \uparrow I_D \downarrow$ 截止 $T \uparrow I \uparrow$
- ### Chapter 3 数字集成电路速度
- ①延时模型 $t_p = 0.69RC$ $t_r = t_f = 2.2RC$ CMOS反相器延时 $t_{PHL} = \frac{C_L V_{swing}/2}{I_{av}} = 0.69 R_{on} C_L$
- $t_p = 0.69 C_L \frac{R_{eq1} R_{eqn}}{2} \Rightarrow R_{eq} \propto \frac{V_{DD}}{I_{DSAT}} \propto \frac{1}{W}$ $V_{DD} \downarrow$ 接近 V_T 时 $R_{eq} \uparrow$ $V_{DD} \uparrow$ V_{DD} 影响大, $V_{DD} \gg V_T + \frac{1}{2} V_{DSAT}$ 无变
- $t_{PHL} \propto \frac{C_L V_{DD}}{I_{DSATn}}$ $t_p \propto \frac{C_L V_{DD}}{k_n (V_{DD} - V_T)^2}$ 漏电容 $C_{gd1} = \alpha W L C_{ox} (L_1)$ 覆盖电容 $C_{gd} = 2C_{ox} W$ 扩散 C_{db}
- 扇出栅电容 $C_{fanout} = 2C_{ox} W_3 L_3 + 2C_{ox} W_4 L_4 + C_{ox} W_4 L_4 \approx C_{ox} L_{drain} (W_3 + W_4)$
- i) 延时 $t_p = 0.69 R_{eq} C_{int} + 0.69 R_{eq} C_{ext} = t_p + t_{effort}$ 4 种负载 负载
- 尺寸优化 $t_{PHL} = t_{PLH} \Leftrightarrow R_{eqn} = R_{eqp} \Rightarrow (\frac{W}{L})_p : (\frac{W}{L})_n = \mu_n : \mu_p \approx 2 : 1$
- 修正 $t_p = t_{p-step} + t_{edge} (1 + 2 \frac{W_1}{V_{DD}}) / 6$
- 设计 $\Rightarrow t_{rmax} = t_{fmax}$
- ### 复杂逻辑门延时
- ①RC树 $T = \sum_{j=1}^N C_i (\sum_{j=1}^N R_j) = RC \frac{N(N+1)}{2} = RC \frac{N+1}{2N} \sim O(N^2)$
- 两管串 $\Rightarrow L = 2L_0$ 并 $\Rightarrow W = 2W_0 \Rightarrow$ 串联后靠近out管 $V_{in} > V_{T0}$ 导通电阻略大 扇入 ≤ 4
- 本征扇入与延时关系 n 扇入 NAM $W_p = 2$ $W_n = 1$; $R_{eq} \sim \frac{1}{n}$ $C \sim nC_0$ 寄生电容 $\sim O(n^2)$
- 本征延时: 尺寸参数 $S = N =$ 驱动提升 $(\frac{W}{L}) : (2)$ $t_p = 0.69 R_{eq} C_{int}$ $R_{eq} \propto \frac{1}{S}$ $C_{int} \propto S$
- 对图 $S = 2 + 2 + 2 = 6$ S 计算和输出 F 直接相连 MOS 尺寸之和
- 尺寸设计并联不变, 串联 n 个, 每个需要 n 倍, $P = 6/3 = S/3 = 2$
- 努力延时: $t_{effort} = 0.69 R_n C_{ext}/S$ 本征时间常数 $\tau = R_n C_n = R_n (2C_{gm} + 2C_{jg}) = 4 R_n C_n$
- τ 计算. 每个输入对应 MOS 尺寸之和 e.g. $A \Rightarrow 2 + 2 = 4$
- 逻辑努力 $g = t_{gate}/\tau_{inv} = t_{gate}/3$ e.g. $g_{2-NAND} = 4/3$
- $t_{effort} = 0.69 \tau \frac{C_{ext}}{C_g} = 0.69 \tau f_r$ 电气努力扇出 $t_{p0} = 0.69 R_n C_{intinv} \Rightarrow t_{effort} = t_{p0} g \frac{f}{f_r}$ $r = 1$

本级延时 1/2-级门延时 门输入

1/2-级门延时 $t_{pgate} = 0.69 R_d C_{int} + 0.69 t_{gate} f = p t_{po} + g \frac{f}{f_0} t_{po} = t_{po} (p + g \frac{f}{f_0})$ $d = p + h/2 = p + g f/h$

eg. $P = 7/3$ 输出Y连接 $4+2+1$; $g_A = (4+2)/3$ $g_B = (4+2)/3$ $g_C = (4+1)/3$



关键路径: 延时最长, 不同信号翻转有不同关键路径

逻辑路径优化 $D = \sum_{i=1}^N (P_i + h_i/f)$ $h_i = g_i f_i$ $F = C/C_g$ $G = g_1 g_2 \dots g_N$ $H = GF$

设每级尺寸参数 S_i , 优化 $S_i \rightarrow D_{min} \Rightarrow$ 最小延时条件 $h = h_1 = h_2 = \dots = h_N$ $D = \sum P_i + \frac{NH^{\frac{1}{N}}}{f}$

eg. 反相器链 $g_i = 1$ $f_i = h/g_i = h = F^{\frac{1}{N}}$ $C_{g_{i,j}} = \sqrt{C_{g_{i,j-1}} C_{g_{i,j+1}}} \Rightarrow$ 逐级放大 $f \Rightarrow t_p = N t_{po} (1 + \sqrt{N}/f)$

分支努力 $f = \frac{C_{out} + C_{eff}}{C_{in}}$ $b = \frac{C_{out} + C_{eff}}{C_{in} - C_{out}}$ $\Rightarrow f/b = C_{in}/C_{out} \Rightarrow F = \prod_{i=1}^N f_i/b_i \Rightarrow$ 有分支路径 $H = GFB$

最优级比: 级数 $f = \exp(1 + 1/f) \Rightarrow f = 4$ 优化缓冲器级数 插入 $N - n_1$ 级 $D = \frac{NH^{\frac{1}{N}}}{f} + N - n_1 + P \Rightarrow \hat{N} = \log_4 H$

求解思路 ① $H = GFB$ ② $N = \log_4 H$ ③ $h = H^{\frac{1}{N}}$ ④ $G_{(min)}/C_g = h/g_i b_i$ $S_i = G_i/g_i C_{ref}$ 反相器放最后.

Chapter 4 功耗

逻辑门功耗 $E_{on} = C_L V_{DD}^2$ $P_{cv} = C_L V_{DD}^2 f_{on} V_{sw}$ 全摆幅 $V_{sw} = 1$ 考虑活性 $P = C_L V_{DD}^2 \alpha_{on} f$

$\alpha_{on} = P_o/P_i = P_o/(1-P_o)$ (独立输入) N 输入静态门 $\alpha_{on} = P_o/P_i = \frac{N_o}{2^N} \cdot \frac{N_i}{2^N} = \frac{N_o(2^N - N_i)}{2^{2N}}$

短路功耗 $P = \frac{1}{2} I_{peak} \cdot V \cdot f$ 翻转 $V_{DD} < V_{tn} + |V_{tp}|$ 完全消除短路功耗

静态功耗: I_{SUB} 亚阈值漏电; I_{GS}, I_{GD} 栅漏电流; I_{LEAK} 漏极漏电. GIDL 影响因素 $V_T, T, V_{DD}, W/L, \mu/L$

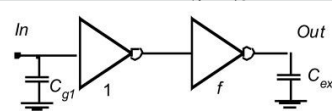
I_{Sub} : 截止器件 $I_{off} = 2 \left(V_{GS} - V_{DS} = V_{DD} = \frac{W}{L} I_{DS0} \left(1 - \frac{(V_{GS} - V_{DD})}{5} \right) (1 - e^{-\frac{V_{GS}}{V_T}}) \right)$ 堆叠效应 V_{DS} 变化 $\Rightarrow I_2 = I_1 (1 - e^{-\frac{V_{GS}}{V_T}})$

功耗与延时关系 $t_p = \frac{C_L V_{DD}}{k_s \mu C (V_{DD} - V_T)^2}$ $P_{sta} = I_{off} V_{DD}$ $PDP = P_{av} \cdot t_p$ eg. inv $P_{av} = C V_{DD}^2 / t_p \therefore PDP = C V_{DD}^2$

$Req \uparrow t_p \uparrow P_{av} \downarrow$; $Req \downarrow t_p \downarrow P_{av} \uparrow \Rightarrow$ trade-off $\Rightarrow EDP = PDP \cdot t_p = P_{av} t_p^2 \Rightarrow EDP \propto ED^{0.5}$

尺寸调整 $C_{load} = C_k \prod_{i=1}^N h_i/g_i$ 电压优化 $V_{DD} \downarrow N$ 级 inv $D = \frac{N C_L V_{DD}}{2 \cdot 0.9} = \frac{N C_L V_{DD}}{2 \cdot f_{ff} 10^{10} s}$ $\Rightarrow E = V_{DD}^2 (C_{eff} + W_{eff} N C_L 10^{-\frac{10}{9}})$

设 $V_{tn} = V_{tp}$ $t_p \approx \frac{2 C_L V_{DD}}{V_{DD} - V_{TE}}$ $V_{TE} = V_T + \frac{1}{2} V_{DSAT} \Rightarrow PDP = \frac{1}{2} C_L V_{DD}^2$ $EDP = \frac{2 C_L^2 V_{DD}^3}{2 (V_{DD} - V_{TE})} \Rightarrow EDP \propto V_{DD}^3$



尺寸, V_{DD} 优化. P_{st}, f, V_{DD} $t_p \leq t_{pref}$ $t_p = t_{po} (1 + \frac{f}{f_0}) + (1 + \frac{f}{f_0})$

$t_{po} \propto \frac{V_{DD}}{V_{DD} - V_{TE}}$ if $r=1$ $t_p/t_{pref} = \frac{V_{DD}}{V_{ref}} \cdot \frac{V_{ref} - V_{TE}}{V_{DD} - V_{TE}} \cdot \frac{2 + f + \frac{f}{F}}{3 + F} = 1$

$E = V_{DD}^2 C_g [(1+r)(1+f) + F]$ $E/E_{ref} = (V_{DD}/V_{ref})^2 (2 + 2f + F/4 + F)$

$f \downarrow C_{eff} \downarrow$; $t_p < t_{pref}$ $V_{DD} \downarrow$ 用性能换功耗